

Karthik Iyer

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PROFESSIONAL SUMMARY

Data Scientist with 2+ years building predictive models, data forecasting, and optimization in Python and SQL on operational and customer data, plus data cleansing, machine learning, and big data (Spark) insights presented to non-technical teams.

EDUCATION

George Washington University — M.S. Data Analytics · GPA 3.68

Washington, DC · May 2025

University of Mumbai — B.E. Computer Engineering · GPA 3.50

Mumbai, India · May 2022

Certifications: Azure ML Pipelines, Google Data Foundations, Google Analytics (GA4)

WORK EXPERIENCE

RestoreFast Operations LLC | Data Systems Engineer, AI and ETL Developer

Bethesda, MD · Jun 2025 – Jul 2026

- Built **predictive models** and reporting in **Python** and **SQL** over operational data integrated from 6 business systems (Gmail, Drive, Xero, Procore, Nutshell, Supabase), giving managers one place to answer client questions.
- Engineered **data cleansing and integration** algorithms in Python ETL pipelines over 40K+ messages and 500+ documents, merging records across systems with no shared customer IDs so each client appeared once, not six times.
- Analyzed **historical patterns in customer and product behaviors** from PostHog usage data across web and mobile apps, finding field staff used only 5–8 of 30 features while admins used all 30, showing how each user group worked.
- Implemented a set-cover **mathematical model** in Python for **operational design** of crew staffing, ranking minimal skill-covering crews by certification, availability, and reliability across 500 staff, so managers staffed jobs in one pass.
- Built an **AI system** in Python and LangChain that turns plain-English questions into SQL over Supabase, so non-technical staff pulled their own project and financial reports instead of waiting on an analyst to write queries.
- Deployed an AI voice intake agent on Twilio and the OpenAI Realtime API converting unstructured inbound calls into structured project records routed across 3 service regions, automating 50% of quarterly intake once transcribed by hand.

George Washington University | Teaching Assistant, Data Analytics

Washington, DC · Sep 2024 – Dec 2024

- Built a **machine learning** grading pipeline in Python routing each question type to the right evaluator: deterministic scoring for numerical work, fine-tuned BERT semantic similarity for free text, across 420 submissions from 42 students.
- Recalibrated similarity thresholds each assignment as new response patterns emerged, moving grading from 10 hours per week of line-by-line review to human-in-the-loop spot checks for the course instructor.
- Surfaced per-assignment **trend analysis** of student performance to guide curriculum decisions.

DPSY and Associates | Analytics Intern

Mumbai, India · Apr 2023 – Jun 2023

- Built **data forecasting** projections of energy consumption against renewable generation using **regression analysis** for an energy client, so leadership could plan capital investment against expected demand rather than estimates.
- Built **data cleansing** and rule-based validation in SQL and Python (Pandas, NumPy) on a banking client's CRM records, screening out ~80% of routine records, flagging loan defaults and partial payments, so auditors saw only exceptions.
- Presented insights through 10+ Tableau and Power BI dashboards and PDF deliverables for banking, retail, and energy clients, turning raw records into figures non-technical leadership could act on.

PROJECTS

Support Ticket Classifier | Python, scikit-learn, NLP, Machine Learning

- Built a **predictive model** in **Python** (scikit-learn, TF-IDF, logistic regression) routing support tickets by type (Incident, Request, Problem, Change) from customer text alone: 79.9% test accuracy, 0.82 macro F1 vs a 40% floor.
- Designed the **experiment** before training: dropped columns added after triage and kept 7,985 families of reworded tickets whole across splits, since a random split overstated accuracy by up to 33 points on 16,338 synthetic Kaggle tickets.
- Tested whether the target itself was learnable; department labels beat their floor by only 9 points while ticket type beat it by 38, so I switched targets instead of tuning on unreliable labels.
- Set an auto-routing threshold from model confidence so 46% of tickets route themselves at 96.4% accuracy, while ambiguous Incident-vs-Problem tickets go to a person.

Large-Scale Customer Review Analysis | SQL, Apache Spark, Databricks, NLP

- Constructed a **data mining framework** of ETL pipelines in SQL and **Apache Spark** on Databricks over 10M+ customer reviews and business records (**big data**), partitioned across 5 business categories for per-category analysis.
- Applied NLP sentiment scoring to 7M+ written reviews to compare what customers wrote with the stars they gave.
- Ran **trend analysis** showing star ratings and written sentiment systematically disagree and that the gap differs by business type, so ratings alone misread satisfaction.